Can high school students think mathematically like Archimedes?
A design research experiment in progress.

A proposal for participation in ICMI-16 from Ralph Mason, U of Manitoba
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A class of academically oriented high school students is offered a problem, and a set of resources they could use to solve the problem. For the use of each resource, however, there is a point-cost which reduces the outcome of the team. Up to 100 points will be awarded for a clear and accurate solution. The resources are:

A tube container just large enough for the three tennis balls it holds: 60 points
A tennis ball: 30 points
A cloth tape measure: 20 points
A metrestick: 10 points
A 1-metre string: 5 points.

Students have ten minutes to decide on the resources they will use, and will then have a further ten minutes to answer the question. The question is about the tube container with three tennis balls. By what percent does its height exceed the distance around the tube (or vice versa)?

In a series of design experiments (Cobb et al, 2003; Lobato, 2003), we have been using this question to introduce a unit of mathematics that engages students in looking more deeply into the formulas for the circumference and area of circles. The students who have participated in the design experiment have learned in previous years that $C = 2 \pi r$, $C = \pi d$, and $A = \pi r^2$, but it has been a repeated source of disappointment for us how few students actually consider using the relationships summarized by the formulas to answer the above question. Comparing circumference with three diameters should be possible without any numbers—thus, without ‘buying’ any of the resources. (The height of the can is three diameters, and its circumference is π diameters, and so the ratio of $\pi / 3$ provides an answer that is worth 100 points.) It appears that the students’ conception of formulas is as recipes for the creation of a desired numerical answer from given numerical values. On one level, the design experiment’s research purpose has been to explore the nature of students’ understandings of mathematical relationships and formulas as it relates to their sense of the nature of the mathematics they learn, while its instructional purpose has been to develop a unit of challenging mathematics that reorients students to actively pursuing richer understandings of mathematics.

However, we perceive a deeper level of pedagogic problematic in the situation described above. We are asking, both generally and within the framing of the design experiment research being described here, *why* the capable students we teach have *chosen* to develop a purely procedural understanding of the mathematical relationships they are studying. We believe that the students do not perceive the potential benefits, including the intrinsic satisfaction, available to them if they accept the challenge of developing richer understandings. It isn’t just about diameters and circumferences or the purpose of formulas. It’s also about the nature of school math, and their participation within it. In the instructional element of this research, we are offering students a greater level of challenge than they are used to in mathematics: the challenge of understanding and expressing in multiple ways the *sense* of the relationships among those formulas, including, for example, *why* the constant value in the circumference and area formulas is the same remarkable value, π. Our challenge as educators is not just to make challenging mathematics available in school. It is to enable, invite, and scaffold students to accept and exploit the challenge such mathematical understandings can offer them (Mason & Janzen Roth, 2004).

Design experiment research (Cobb et al., 2003; Shavelson et al., 2003) is a focused form of action research. Its structure involves the development of instructional resources and strategies for potential implementation beyond the scope of the research itself. However, its purpose as a form of educational research is to explore the qualities of student understandings and their development of further understanding as the instruction progresses through its cycles of design, implementation, and evaluation. In this case, the instruction deliberately and aggressively attempts to reorient students' approaches to learning mathematics toward the challenge of developing conceptual understandings rather than the stockpiling of additional procedures, and it uses challenging mathematical activities drawn from an academic study of the history of mathematics to accomplish its educational goals. Our intention is to have a third full cycle of design experimentation completed by June 2006. We will have instructional resources ready to recommend for broader use. There will be research outcomes regarding the feasibility of challenging mathematics inquiries for reorienting students toward richer forms of learning mathematics.

The history of mathematics as a source of instructional inquiries

The curriculum design for this teaching experiment began with a study of the history of mathematicians' inquiries into circle relationships, especially Archimedes (Cuomo, 2000; Eves, 1960) and early Chinese mathematicians (Liu, 2003). There are a variety of reasons for the use of the history of mathematics as a source of mathematical activity for classroom use. One, it is our responsibility to offer students an opportunity to understand mathematics as more than a static body of knowledge, as an ongoing process, a part of our culture, developing through thoughtful effort our communal mathematical understandings (Arons, 1991; Mason, 1999). Two, historical framings of mathematics provide a way to put a human face on the mathematics that students encounter, offering context and story-lines to enliven the content and role/process models for students to view as examples (positive and negative) for their own mathematical efforts (Stinner & Williams, 1998; Mason, 2003b). Three, although individuals' understandings do *not* necessarily follow the historical order in which mathematics developed, the historical structure of the discipline offers a framework within which educators can think about the educational sequence and structure of topics (Rudge & Howe, 2004; Mason, 2001). No matter what the theoretical reasons, however, it is now necessary for those of us who are promoting the sincere use of the history of the discipline to demonstrate its actual value in educational terms (Lederman, 2003; Mason, 2003a). This work contributes to that goal, in part by demonstrating the possibilities of offering more challenging, more rewarding mathematics to students in their school mathematics.

It is important to outline two premises of our work with the history of mathematics which may differ from readers' anticipation of such work. First, we have not accepted the historical record of mathematics as a linear sequence of formal results. We recognize and understand the utility of mathematicians presenting their mathematics only in final form, with fully developed deductive reasoning as support (Hoffman, 1988; Mason, 2004) but without a record of the processes by which the mathematical inquiry itself progressed. However, we also recognize that mathematics documents, by their very nature as precise and fully grounded works of mathematics, obscure the mathematical cognition that preceded them, and we must be diligent in our search for the mathematical processes that *led* to the ideas in such documents. There are, of course, incomplete records of such processes. The discovery in the early 20th century of a partial copy of Archimedes' *The Method* provides an example of documents that offer glimpses into the thought processes that precede the publication of mathematicians' formal results. Other aspects of those thought processes need to be inferred from considerations of the historical record of the lives and societies in which the mathematicians did their work (Joseph, 2000).

Second, we have not looked to the historical development of mathematics for processes for students to replicate. On a practical level, it's not easy to sort out the apocryphal accounts of mathematicians and scientists from the plausible or likely accounts. In fact, our desire for a good quick story may have favored the apocryphal anecdote over more detailed and accurate accounts of mathematicians' cognition (Fara, 2002; Mason, 2003). Two examples are the Eureka story of Archimedes and the bathtub, and Newton being struck (literally) by gravity when an apple fell on his head. We cannot afford to view or represent mathematical cognition as bright-flash happenstance moments if we want to cause students to accept the challenge of doggedly but creatively constructing richer understandings of the objects of their attention.

There is a much more important reason, however, for moving beyond replication. Although much can be used from replicating the actual processes by which mathematical and scientific premises were conceived and tested (Heering, 2005; Mason & Stinner, 2001), it is neither practical or purposeful within the broader educational endeavor to have students using the tools available to Archimedes (papyrus or sheepskin and ink, sand-tables, clay tablets with a stylus, for instance). Not only do we want students developing capability with the tools of today for pragmatic reasons (National Council of Teachers of Mathematics (NCTM, 2000). It is also in the spirit of the historical record that mathematicians' tools of choice were the tools available to them in the time and place, the culture, where they lived. But most important, we must recognize that our purpose in a classroom is not to relive history, in this case the brilliant development by a single mind of a signal idea. Our purpose is educational, the development by as many students as possible of as rich and as interwoven a conceptual understanding as possible. We must avail ourselves of the most potent supports for student learning as possible, and that means moving beyond historical replication to historically grounded instructional design of activities using today's learning tools (Mason & Janzen Roth, 2005).

In brief, we use a rigorous analysis of the history of mathematics to develop a plausible conception of the cognition that led to the mathematics that our students learn. We then adapt the processes of inquiry that led to the development of that mathematics into curriculum that enables students to inquire into the same mathematics. The challenge for the students is in part the mathematics content—to develop a rich conceptual understanding of the mathematical relationships behind the formulas (e.g, the concurrent precision and irrationality of the constant in the formulas for the circumference and area of circles). The challenge is also to appreciate and develop one's skills in mathematics as inquiry, including mathematical thinking (Huckstep,2002; J. Mason, Burton, & Stacey, 1982), mathematical problem posing (Brown & Walter, 1983), and mathematical problem solving (Polya, 1960; NCTM, 2000). The history of mathematics gives us the mathematical version of the inquiry processes behind the content. Design experiment research may give us the mechanism, over time, to develop instructional versions of those processes which preserve the spirit, challenge, and intrinsic rewards of the mathematician's original inquiries.

The instructional processes

Details of the instructional processes in our design experiment continue to be redeveloped as the experiment continues. However, the unit in its current design consists of about ten hours of instruction. It consists of three general strands. Mathematical inquiries into the relationships among quantities of the circle are the primary strand. Attached to that strand are processes of interaction, oral and written, about the relationships and the processes within the inquiries. Third is a sequence of anecdotes with associated student activities that bring Archimedes into the students' imagination.

Although there are more in the actual unit, three mathematical inquiries may portray the quality and range of these curriculum elements. In one of the inquiries, students inscribe and circumscribe a series of regular polygons within circles, measuring the edges and apothems and calculating lower and upper bounds for the circumference of the circle. The activity stops short of Archimedes' method of exhaustion for determining circumference to any desired degree of accuracy, but the activity gives students the opportunity to (first) construct and think about circles and polygons without using measured lengths and angles and (then) use measurements to confirm the progression in the relationships among the lengths. In another inquiry, students build "radius squares" to match a set of progressively larger circles, and graph the quadratic relationships of the edge of each square (the radius) with the areas of both the square and the circle. They then graph the relationship between the area of the squares and the area of the circles, and discover it to be, counter to their intuition, to be linear. In another activity, the students cut circles into equal sectors and rearrange the sectors within a rectangle whose base is the circumference and height is the radius. They repeat the process, with twice as many sectors each of half the size; they repeat it a third time, now with sectors much smaller than the sections of an orange-slice. In the manner of a thought experiment (Winchester, 1991), and approximating closely the thinking of Archimedes that foreshadowed the calculus of Newton, the students have spontaneously extended the process, imaging sectors that are infinitesimal in width; the students can readily see that the sectors would exactly half-fill the rectangle. They can 'see' (and express in words and symbols) the relationship between the linear construct, circumference, and the two-dimensional construct, area; and they see the presence of the same constant, π, in both. (The activity deserves to be shared as a process, rather than summarized in the few sentences used here.) Each activity contributes to the experiences, the imagery, and the communicative competence of the students. It is not the presumption of the instructional design, however, that the challenge to understand a specific formula in a specific way is represented by a specific mathematical inquiry. Rather, it is the generalization and abstraction of experiences from the group of inquiries overall that seems to be leading students to accept the challenge and succeed generally to understand the mathematics of circles as Archimedes did.

The second element of the design is the communicative element. Communicative processes include the use of *groups of four* (Johnson & Johnson, 2004) for collaboration when the whole class does the same inquiry. With other inquiries, half of each group does a different inquiry, followed by pairs teaching pairs the processes and outcomes they experienced. The students' written work for each inquiry includes a piece of interactive writing (Mason & McFeetors, 2002), giving the student the chance to think about and express their own mathematical thinking or values. As well, the teacher's personal responses help the student to recognize the teachers' purposes and concern for their achievement. Closure interviews with each person used narrative inquiry methods (Clandinin & Connelly, 2000; McFeetors & Mason, 2005) to foster students' retrospective thinking about the unit when it was completed. In part, communicative elements provide the teacher-researchers with the data that the research depends upon. However, the communication processes are essential aspects of the processes that lead students to upgrade their engagement with the mathematics beyond minimal computational competence.

The third element brings the historical record into the students' view. It consists of short verbatim anecdotes, taken from ancient (e.g., Cicero) and modern (e.g., Barrow, 1992) versions of Archimedes' life. Each anecdote comes with a demonstration or an activity that enables the students to be intellectually active with the anecdote. For example, after the Sand Reckoner story, students are led to use exponential representations to calculate the grains of poppy-seeds that would fit in a cubic metre, after estimating the grains that would fit in a centimeter cube. To date, we have made progress with keeping the anecdotes and activities from expanding beyond reason the time the unit takes. We need to make more progress with making the historical-record element as effective overall as the other two strands have been.

Research results to date

The research continues, and as a result I will present only a brief description of the results to date. First, we have found that a wide range of beliefs and values about the nature of academic mathematics and its rightful purpose in school. Efforts to reform students' orientations toward school mathematics will need to be mindful that students' prior beliefs vary widely but appear to be persistent in the face of invitations to change. When probed, students have portrayed their beliefs as deeply embedded in their lived histories, products of their personal experiences and their social environments. Second, we have found that students' understanding of functions and relationships and the formulas that (to mathematicians) summarize those relationships are disappointingly shallow. Yet, the shallowness seems to be remediable, through engaging students in challenging mathematical inquiries related to those relationships. Third, we have found that the educational research to date on students' understandings of algebra—its functions, its multiple representations, its component concepts such as variable and slope—have lots to say in guiding design experiment activity such as ours, and may provide further illumination in turn. This aspect of our results, however, should not be represented briefly or prematurely.

The relationship to ICMI-16.

I would like to believe that this work has much to offer. Those who know me from my past work know that I value highly the invitational qualities of the mathematics that we offer to students (Mason, 1998; Mason, 2002; Mason et al, 2005). In some ways, this work is a both an idealistic and pragmatic extension of that work. It has been rewarding to perceive the history of mathematics as an informative source not only of possibilities for classroom inquiries, but also for the development of a more coherent image of the challenge of mathematical inquiry. Translation of lived mathematics into classroom activity is a challenge for us as a community of scholars, and these early attempts offer some promise while also making certain limitations and challenges more apparent. The design experiment research method may well offer to the working group a promising approach to answering (over time) those questions associated with bringing challenging mathematics into classrooms. Compared to other proposals, I think this line of activity offers two powerful working ideas:

1. Mathematics is powerful because it is inherently challenging. That power is not only a matter of the potency of application or the provision of foundations for further mathematical inquiry. It is also the potency that comes from engaging in (and succeeding at) rich complex activity and the construction of robust images and representations of that activity and its contents. What this project may represent at ICMI-16 is that we may be able to make the power of challenging mathematics an inherent part of students' experiences of school mathematics in ways that can include and reward the broad range of students now participating in school math. Along with the extracurricular or special-audience possibilities listed by the group's organizers, this work suggests that perhaps *challenging mathematics can be offered within classrooms*, within core content, within existing curriculum. Students (and teachers) can be convinced to accept the challenge as they recognize the rewards inherent in instruction organized around the challenge of mathematical thinking.

2. Design experiment research may provide a way to bridge the relationship between mathematics and pedagogy, mathematicians and teachers. My work in this area has been a full partnership with a remarkable young teacher, and he and I have found our work to be accessible and valuable to members of the three communities we represent: mathematics teachers, mathematics educators, and mathematicians. Design experiment research may offer ways for our community to explore our conceptual inquiries while we develop our instructional offerings for others to use. It is important to note here that although design experiment research can operate well on large-scale endeavors (Cobb et al, 2003), my partner and I have been keeping it small, dealing with a single (though broad) topic, developing a single unit of instruction, and addressing a focused research agenda. ICMI-16 members may appreciate its possibilities for incorporating its methods or principles into their ongoing research into challenging mathematics instruction.

However, my main reason for hoping to participate in ICMI-16 is not because I think this work has something to offer to the scholarly community being formed. It is because I am sure the community will have lots to offer in relation to this work. This work needs to be situated and informed, critiqued and redirected, by a larger community of thinkers. And on a personal level, I recognize a need to be engaged in a broader discourse about the nature of mathematics for school.

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